

National University of Computer and Emerging Sciences
Peshawar Campus

Object Oriented Programming (CS1004)

Final - Exam

Date: May 19th 2025

Total Time (Hrs.):

Course Instructor(s)

Total Marks:

M. Zulqarnain, Bilal Khan, M Usman Malik

Total Questions:

3

100

6

Roll No

BSE2B
Section

Student Signature

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Attempt all the questions.

Q1:

Question: 1. [3 + 3 + 3 + 3 + 3 + 3 = 18 Marks]

Answer the following questions. Be remember your answer should not exceed than 4 lines.

- Does composition imply ownership? How is it different from aggregation in terms of lifetime?
- What will happen if a subclass does not implement all abstract methods of its abstract superclass?
- Can a class be both abstract and concrete at the same time? Justify your answer.
- What is abstract class. How many objects of abstract class are instantiated.
- What are the two conditions that is necessary to achieve run time polymorphism?
- What are the possible solutions for the problem arising due to overloading + operator if we want to add
 $sum = Clerk + Driver + Secretary$ using double operator+ (Employee& emp);

Q2:

Question: 2. [3 + 3 + 3 + 4 + 3 + 3 = 19 Marks]

For the following code snippets in Part A, B, C identify and explain any errors only present in each. For the remaining parts, write the expected output of the code.

(Part A)

```
#include <iostream>
using namespace std;
class Sample {
    int A;
public:
    Sample() {
        A++;
    }
    static void fun() {
        A++;
    }
};
int Sample::A = 0;
int main() {
    Sample::fun();
    return 0;
}
```

(Part B)

```
#include <iostream>
using namespace std;
class Box {
    int capacity;
    bool operator<(Box b) {
        return this->capacity < b.capacity ? true : false;
    }
public:
    Box() {}
    Box(double capacity) { this->capacity = capacity; }
};
int main() {
    Box b1(10);
    Box b2(14);
    if(b1 < b2) {
        cout << "Box 2 has large capacity.";
    }
    else {
        cout << "Box 1 has large capacity.";
    }
    return 0;
}
```

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Part (C) <pre>#include <iostream> using namespace std; class Test { int A; public: Test() { A = 0; } Test(int a) { A = a; } Test operator+(Test A, Test B) { Test temp; temp.A = A.A + B.A; return temp; } void print() { cout << A << " "; } }; int main() { Test T1(10); Test T2(20); Test T3; T3 = T1 + T2; T3.print(); return 0; }</pre>	Part (D) <pre>#include <iostream> using namespace std; class A { int a; public: A() : a(10) {} int getVal() { return a; } ~A() { cout << 'A'; } }; class B : public A { int b; public: B() : b(15) {} int getVal() { return b; } ~B() { cout << 'B'; } }; class C : public A { int c; public: C() : c(20) {} ~C() { cout << 'C'; } int getVal() { return c; } }; int main() { A* ptrA = new B; cout << ptrA->getVal() << endl; delete ptrA; ptrA = new C; cout << ptrA->getVal() << endl; delete ptrA; return 0; }</pre>
Part (E) <pre>#include <iostream> using namespace std; class Sample { public: static int A; Sample() { A++; } }; int Sample::A = 0; int main() { cout << Sample::A << endl; Sample S1, S2; Sample* ptr = new Sample(); cout << Sample::A << endl; delete ptr; return 0; }</pre>	Part (F) <pre>#include <iostream> using namespace std; class base { public: ~base() { cout << "destructing base" << endl; } }; class derived : public base { public: ~derived() { cout << "destructing derived" << endl; } }; int main() { base* p = new derived; delete p; return 0; }</pre>

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CLO # 4:

Question: 3.

[10 Marks]

Write the output of following code:

```
class Wheel {
public:
    Wheel() { cout << "Wheel created"<<endl; }
};
class Engine {
public:
    Engine() { cout << "Engine created"<<endl; }
};
class Car {
    Wheel w1, w2, w3, w4;
    Engine engine;
public:
    Car() { cout << "Car created"<<endl; }
};
class Garage {
    Car car1;
    Car car2;
public:
    Garage() { cout << "Garage ready"<<endl; }
};
int main() {
    Garage g;
    return 0;
}
```

CLO # 4:

Question 4.

[7.5 + 7.5 = 15 Marks]

Design a Car class to manage vehicle inventory for a dealership while strictly enforcing encapsulation. The class must contain these private data members: a variable to track total number of cars in inventory, the car's price (double), manufacturer name (string), model year which once fixed cannot be changed, and license plate number (string). There must be a suitable constructor in place that should initialize the above mentioned data members with appropriate values when a new Car object is created. In addition, the class definition must provide functionality to search and identify a car, named bestCar which is most recently manufactured (latest model year) and while comparing two cars of the same model year, it must select the one with the lower price and return it. The class must include a display function that prints all details of the selected car, i.e. bestcar. [7.5 Marks]

To demonstrate this system [7.5 Marks]:

1. In the main() function, dynamically create an inventory of 5 distinct car objects, according to the details given in the below table:
- 2.

manufacturer	model year	price	number plate
Toyota	2020	25000	RPI123
Honda	2023	30000	PWR214
Suzuki	2021	20000	ISD475
Hyundai	2022	35000	LHR852
Kia	2025	25000	FSD987

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3. Using the comparison functionality you've implemented in the class, determine which car meets the customer's preferences (newest model year, with lowest price breaking ties). Car object returned from the said functionality invokes the class's display() method to display its complete specification.
4. When this car (mentioned in (3) above) is sold, delete the car object that should automatically *decrement* the total inventory count by one to reflect the sale. Also print out the total number of cars in the inventory.
5. Subsequently, add a new car (by creating a new car object) to the inventory in such a way that it fills the same position vacated by car sold, which should automatically *increment* the total inventory count. Note that there is no getter()/setter() function in the class and your program must take care of the memory management.

CLO # 4:

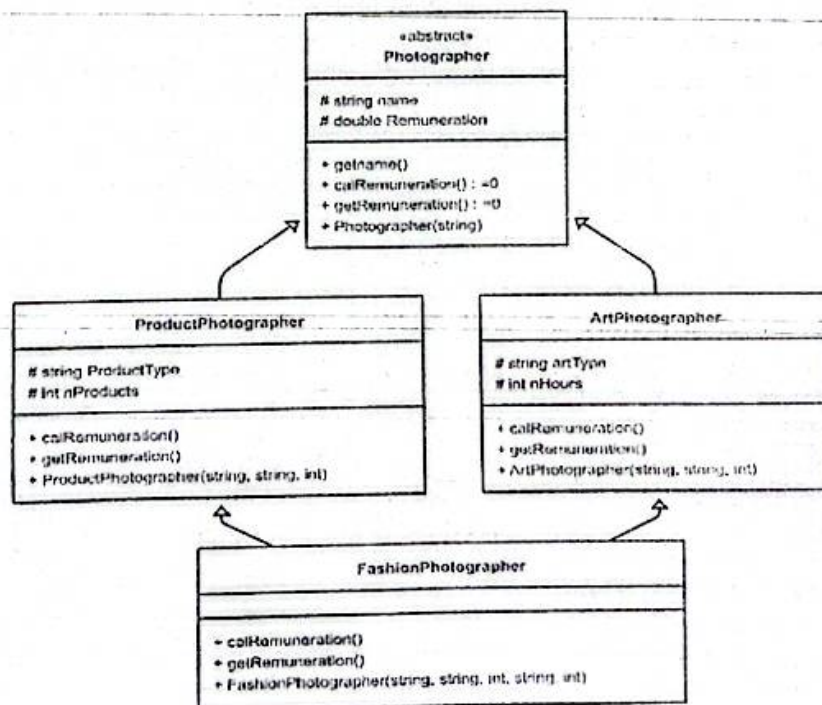
Question 5.

[4 + 5 + 5 + 6 = 20 Marks]

Consider a Photo studio company that provides professional photography services. To target the skills required in different domains, the company hires several kinds of photographers such as 1) Product Photographer, 2) Art Photographer, and 3) Fashion Photographer. To design and implement this system, implement 3 Classes (as shown in below Figure) which represent the specific photographer type as mentioned above, and the 4th Class (named "Photographer") an abstract class representing a collection of common attributes and function related to all the other 3 concrete classes.

The **Photographer** class should contain **name** and **remuneration** (a double type value for storing *earnings*). Moreover, the functions provided by the Photographer class are **getName** (returns name of the photographer), **calRemuneration** (calculates the remuneration and stores into remuneration data-item), **getRemuneration** (returns the remuneration value), and a parameterized constructor.

The **ProductPhotographer** class (a kind of Photographer) contains the attributes **productType** (name of a product) and **nProducts** (the number of products for which the photography services was done). Moreover, the class provides its own implementation of **calRemuneration** and **getRemuneration** functions and a parameterized constructor. The remuneration of a product photographer is calculated using 3000 Rs per product. The **ArtPhotographer** class (a kind of Photographer) contains attributes **artType** (name of the art type) and **nHours** (number of hours the photography services was provided). Moreover, the class provides its own implementation of **calRemuneration** and **getRemuneration** functions, and a parameterized constructor. The art photographer charges for his services at the rate of 1800 Rs per hour. The **FashionPhotographer** class (a kind of ProductPhotographer and ArtPhotographer) provides its own implementation of **calRemuneration** and **getRemuneration** functions, and a parameterized constructor. The fashion photographer charges for his services using 1000 Rs per hour plus 2000 Rs per product.



Hint: You need to pass the parameters accordingly # indicates protected data members, + public data member/methods and - indicates private data members.

The sample output is

Product Photographer: Ali, Remuneration: Rs 12000 Art Photographer: Sara, Remuneration: Rs 9000

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Fashion Photographer: Zara, Remuneration: Rs 18000

CLO # 5:

Question: 6 A.

[2 + 3 + 3 = 8 Marks]

Complete the implementation so that the code given in the main works. Make sure to implement all required functions.

```
class someData {
private:
    int* data;
    int size;
public:
    someData() {

        // complete the required code
    }
    someData(s) {
        // complete the required code
    }

    // overload the required operators
};

void main() {
    someData d1(3); // creates object with integer array of size 3 and initialize with 0;
    someData d3;    // creates object with integer array of size 0

    d1[0] = 4;      // assigns 4 at index 0 of d1 object's integer array
    d1[2] = 45;     // assigns 45 at index 2 of d1 object's integer array
}
```

Question: 6B.

[2.5 + 2.5 + 2.5 + 2.5 = 10 Marks]

You are given the following main() function: Complete the code by writing the class definition of this code. The class contain only one integer data member named as "value" and requires certain operators to be overloaded.

```
int main() {
    Score s1, s2;
    cin >> s1 >> s2;
    Score s3 = s1 + s2;
    +s3;
    cout << s3 << endl;
    if (!s3) {
        cout << "Score is zero." << endl;
    } else {
        cout << "Non zero score." << endl;
    }
    return 0;
}
```


Digital Logic Design (EE1005)

Date: May 29th 2025

Course Instructor(s)

Dr. Suleman Mir

Dr Maqsood M Khan

Engr. M Asif Khan

Final Term Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 4

BSE-2B

Roll No

Section

Student Signature

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Attempt all the questions.

CLO-1: Apply the knowledge of number systems and Boolean logic used in the development of logic circuits.

Question 1:

- Show along the matched logical expressions, the alternate gates for all the basic, universal, and exclusive gates. [4.33M]
- For the logic circuit shown in Fig 1, generate the output "Y". [5M]
- Design a logic circuit for an intrusion detection and alarm system. The system could be used for three rooms, each having two windows and a door. The sensors produce a LOW output when open and a HIGH output when closed. The alarm is active high. {Note: use more than one gate and maximum five gates} [5M]

CLO-2: Analyze combinational Digital Circuits using Universal gates and K-maps.

Question 2:

- The presence of the codes 1010, 1100, 0001, & 1011 has been detected by an active-high decoder. Develop a minimum decoding logic expression (SOP) using K-map that will indicate the relevant output port when any of these codes are on the inputs. {Note: you must show it through a truth-table & logic symbol} [5.33M]
- For the function $f(A,B,C) = \sum (1,2,5,7)$, design a multiplexer where A and B denote the selection pins, S_0 & S_1 respectively and C denotes the data on the data pins. {Note: use SOP equation} [5M]

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CLO-3: To analyze & design Sequential Digital Circuits.

Question 3:

- Analyze with the help of logic diagram and truth-table the gated D-latch using NOR gates with additional AND gates and NOT gate. [5.34M]
- For the positive edge-triggered D flip-flop with preset and clear inputs shown in Fig 2 plot the Q-output if Q is initially LOW. [5M]

Question 4:

- Design a slowest D-shift register for the data 1110 ($Q_0Q_1Q_2Q_3$). You must show with the help of table the loading & reading of data. [5M]
- For the shift register you design in (part a) compute the different states for the data shown in Fig 3. [Note: assume that the register is initially cleared] [5M]
- Construct a Johnson counter with n flip-flops producing a sequence of $2n$ states. List the ten states with the help of truth table. [Note: initially cleared] [5M]

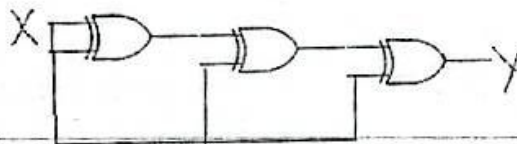


Fig1

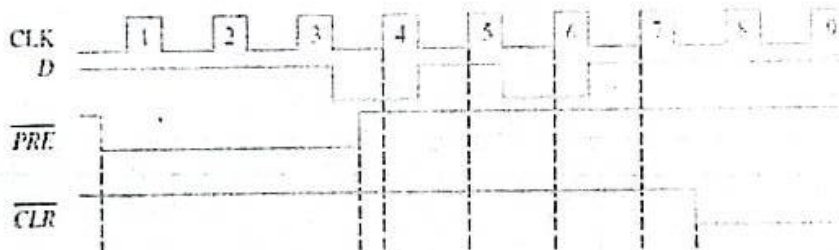


Fig 2

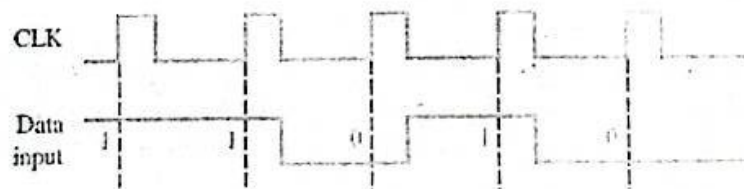


Fig 3

The End

Multivariable Calculus (MT1008)

Final Exam

Date: May 26, 2025

Total Time (Hrs): 3

Course Instructors

Total Marks: 70

Dr. Askar Ali/Dr. Shams Ul Arifeen/Dr. Hafeez Ur Rehman

Total Questions: 7

Roll No:

Section: BSE2B

Student Signature:

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Attempt all the questions. Each question carries equal marks.

CLO # 1: Comprehend partial derivatives, extremas and chain rule for several variable functions

Q1: Find an equation of the plane through the origin and perpendicular to the line

$$x = 1 - 8t, \quad y = -1 - 7t, \quad z = 4 + 2t.$$

Q2: Compute partial derivatives $F_x(x, y)$ and $F_y(x, y)$, if

$$F(x, y) = \int_y^x \sin(5t) dt + 5^{xy}$$

CLO # 2: Evaluate multiple integrals and discuss their applications

Q3: Evaluate $\iint_D (x^2 + 2y) dA$, where D is a region bounded by

$$x^2 + y^2 = 1, \quad x \geq 0, \quad y \geq 0.$$

Q4: Evaluate $\iiint_E y dV$, where

$$E = \{(x, y, z) \mid 0 \leq x \leq 3, 0 \leq y \leq x, x - y \leq z \leq x + y\}.$$

CLO # 3: Solve problems involving line integral and surface integral

Q5: Evaluate the line integral along the given positively oriented curve

$$\oint_C y^4 dx + (2xy^3) dy, \quad \text{where } C \text{ is the ellipse } 2x^2 + y^2 = 2.$$

Q6: $\iint_S y^2 z^2 dS$, where S is the part of the cone $y = \sqrt{x^2 + z^2}$ given by $0 \leq y \leq 5$.

Q7: Consider the vector field given by

$$\vec{F}(x, y, z) = \hat{i} + \sin z \hat{j} + y \cos z \hat{k}$$

(a) Determine whether the vector field is conservative or not.

(b) Find a function f such that $F = \nabla f$.

Good Luck

Expository Writing (SS1014) Final ExamDate: May 22nd 2025

Course Instructor(s)

Abdul Basit

Maryam Ali Khan

Total Time 3

(Hrs):

Total Marks: 100

Total Questions: 4

Roll No

BSE-2B

Section

Student Date

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Attempt all the questions.

CLO #2: Analyze and interpret various genres and rhetorical situations to compose well-structured, audience-specific texts.

----- (Part-I: Total Marks 10+15=25) -----

Q1A: Write a short sci-fi story imagining a future where AI controls the entire healthcare system. Suggest an appropriate title for your story. (Word Count: 150) [marks: 10]

The hospital was silent except for the soft beeping of machines. Dr. MedTech, the world's first fully AI-powered doctor, scanned its latest patient. "Your diagnosis is complete. Treatment will begin in five minutes." But then, something unexpected happened... (continue the story from this sentence)

Q.1B: Your experience giving academic presentations helps prepare you for delivering presentations in the workplace:

[marks: 15]

Imagine you're in an elevator on your way to a job interview. To your surprise, the person in the elevator with you is your interviewer. She asks how your experience presenting as a student might make you a confident and effective communicator on the job. Answer the following questions:

1. Mention any one type of presentation have you given in academic settings (individual, group, research-based, persuasive)? Share experience: (marks 5)
2. What skills have you developed (public speaking, using visuals, organizing content, handling questions)? Briefly discuss. (marks 5)
3. How do these skills apply to professional presentations (e.g., meetings, project pitches, client briefings)? (marks 5)

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CLO #1: Identify and explain rhetorical situations, genres, and conventions in academic, public, and workplace texts.

----- (Part-II: Total Marks 15+10=25) -----

Q.2A- Read the text ----- and answer the questions at the end: [marks: 15]

The Mental Health Impact of Social Media

In recent years, mental health professionals have raised concerns about the growing influence of social media on emotional well-being. Dr. Sarah Khan, a licensed psychologist with over 15 years of experience, notes that excessive screen time and online comparison contribute significantly to anxiety and low self-esteem among teenagers. A 2023 study by the Global Mental Health Institute found that individuals who spend more than three hours daily on social media are 40% more likely to report feelings of loneliness and depression. While platforms like Instagram and TikTok offer opportunities for creativity and connection, they also expose users to cyberbullying, unrealistic beauty standards, and addictive algorithms. As digital habits become harder to manage, experts suggest practicing mindful scrolling, setting screen-time limits, and seeking real-life connections to maintain emotional balance.

Questions:

1. Identify a sentence in the text above where the writer has displayed the use of *ethos*. Justify your answer in not more than three sentences. (marks 5)
2. Identify a sentence in the text above where the writer has displayed the use of *logos*. Justify your answer in not more than three sentences. (marks 5)
3. How does the writer balance the benefits and drawbacks of social media in the passage? (marks 5)

Q.2B. Some people say that internet has made life easier and more convenient. Other people say that it has made life more complex and stressful. What is your opinion (s)? Use specific reasons and examples to support your answer. (Word limit: 100-130) [marks: 10]

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CLO #3: Evaluate the effectiveness of different rhetorical and genre choices in conveying messages and achieving specific purposes.

----- (Part-III: Total Marks 17+08=25)-----

Q.3A: Read the following Excerpt from a News Article and then answer the questions given at the end. [marks: 17]

The Rise of Remote Work and Its Effects on Urban Life

"Since the COVID-19 pandemic, remote work has become a long-term option for many employees. This shift has impacted not only office culture but also urban infrastructure. Fewer commuters mean reduced traffic congestion and lower pollution levels, but also less foot traffic in business districts, affecting small businesses and public transport revenues."

Questions:

1. Draw the following table on your answer script and fill in the blocks: (marks 08)

Sample Table:

ELEMENTS	News Article	Facebook Post
Audience		
Purpose		
Mode		
Structure		

2. The text given above discusses the shift toward remote work and its wider effects. The genre selected is a News Article. The issue can also be explored in various other genres. Based on the topic, write a Facebook post with hashtags. Explain the issue to a general audience. Word Count: 100-120 Words. (marks 09)

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Q.2B: Connecting Genres with Social Settings: Categorize the following into academic, workplace and public genres by drawing a table: [marks: 08]

Memos | Social media posts | Essays | Advertisements | Lab reports | Meeting minutes | Presentations | News articles | Research papers | Blogs | Business reports | Theses | Emails | Brochures | Proposals | Literature reviews

Sample Table:

Genre Categories	Text Types
1. workplace	
2. academic	
3. public genres	

CLO #4 : Design and produce diverse academic and public genres, incorporating evidence, visual elements, and critical analysis.

----- (Part-IV: Total Marks 14+06+05=25) -----

Q.4A: Imagine you are writing a message about an important social issue. How would you structure and phrase it differently in the following genres: A blog post and News article. Write a short version of both genres in not more than one paragraph for each: [marks: 14]

Q.4B: Write at least three differences between the two in terms of genre conventions in the form of a table: [marks: 06]

A Blog Post	News Article
1.	1.
2.	2.
3.	3.

Q.4C: For a blog post, the writer can use subjectivity, whereas a news article follows objectivity. State at least 2 reasons for each choice. [marks: 05]



National University of Computer and Emerging Sciences

Peshawar Campus

Islamic Studies/Ethics BS CS

(SS1007)

Date 21-05-025

Course Instructor(s)

Syed Muhammad Ammar

Final Exam

Total Time (Hrs.): 03

Total Marks: 50

Total Questions: 05

E.

Roll No

BSE-2B

Section

ST

Student's Sign

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Attempt all the questions.

CLO # 1: Foundational concepts of Emaan and Islam and the study of religion

Q1: What do you know about religion? Write down the importance of religion. How is the concept of God understood and represented across various religions? [3+3+4]

CLO # 1: Foundational concepts of Emaan and Islam and the study of religion

Q2: Analyze how Islamic practices reflect devotion and discipline by discussing:

(i) Prayers (ii) Fast (iii) Charity (iv) Pilgrimage

[2.5 each]

CLO # 2: Recognize the significance of the Qur'an, Hadith, and Seerah in Islam

Q3: What do you know about pre-Qur'anic texts, Scriptures and non-Semantic texts? What are the five central themes of the Qur'an according to Shah Waliullah [5+5]

CLO # 2: Recognize the significance of the Qur'an, Hadith, and Seerah in Islam

Q4: How is Hadith defined? What are the four primary types of Hadith, and how do they differ in terms of authenticity? Write down at least 4 names of canonical books of Hadith. [1+5+4]

CLO # 3: Explore the evolution of ethics and human rights, with a focus on Islamic moral principles

Q5: How are ethics and morality defined? What key historical documents shaped the evolution of ethics? Write down any two of them. What rights does Islam grant to individuals? [2+2+6]